# Yi Yuan

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Male | Ph.D. Candidate in Physics, Tsinghua University

## Education

**Tsinghua University Ph.D. in Physics** 2023.08 – 2028.06 (Expected)
Department of Physics Ranked **1st** in Modern Physics in the Ph.D. Qualifying Exam Principles and Architecture of Quantum Computers (A+) Beijing

**Tsinghua University B.S. in Fundamental Sciences, Mathematics and Physics** 2019.08 – 2023.06
Department of Physics GPA: 3.89/4.0 (Rank: 18/116) Outstanding Graduate Nominee, Ye Qisun Award Beijing

## Technical Skills

- **Programming Languages:** Python (primary language for daily research and development)
- **AI/ML:** Familiar with Transformer architectures and RLHF workflows (PPO/GRPO); experienced with JAX and PyTorch; hands-on experience in CNN modeling and LLM API application development
- **Data and Computing:** NumPy vectorized computing, multiprocessing, data-driven modeling, and backtesting system development
- **Tools:** Git, Linux servers **Language:** English – proficient in reading and writing scientific literature

## Publications

**Yi Yuan**, Yuanchen Zhao, and Dong E. Liu. "Zeno-Enhanced Probabilistic Error Cancellation with Quantum Error Detection Codes." *arXiv preprint arXiv:2605.12149*, 2026. arXiv
Developed a perturbative QEDC+PEC protocol for near-term noisy quantum devices, using post-selection to reshape the effective noise channel and PEC to cancel residual logical errors.

## Research and Projects

**Research in Quantum Computing and Quantum Information** 2022.09 – Present
Department of Physics, Tsinghua University Python, JAX, Tensor Network

- **Zeno-enhanced QEDC+PEC protocol** (2024 – 2026, arXiv preprint): Combined quantum error-detecting codes with probabilistic error cancellation for near-term noisy quantum devices; in 200-qubit Iceberg-code logical GHZ preparation, reduced sampling overhead by **3–4 orders of magnitude** relative to standard PEC while maintaining $F \simeq 0.956$
- **Classical Shadow error mitigation** (2024): Used tensor network methods to compute the error-mitigation complexity of classical shadows on noisy quantum circuits with $\log(n)$ depth, and analyzed its scaling with system size
- **Undergraduate thesis: complexity of quantum annealing** (2022.09 – 2023.06): Based on the imaginary-time spacetime instanton action, explained why quantum annealing and classical QMC have matching complexity in ground-state energy search for double-well potentials, and identified the possibility of multi-path quantum speedup

**Selected Projects**

- **SEVI particle scattering analysis**: Built a PyTorch-based CNN to fit scattering-amplitude parameters, reaching the accuracy of analytical methods without injecting prior knowledge
- **LLM paper summarization tool** (2025): Automatically fetched daily arXiv papers, generated summaries through the Kimi API, and completed local deployment

## Honors and Leadership

Honors: 2021 Tsinghua Alumni – Zhang Mingwei Scholarship Multiple Department of Physics scholarships for academic excellence, innovation, and overall achievement in 2020–2021
Leadership: 2023.08–2025.01 Student Counselor, Department of Physics, Tsinghua University 2019–2020 Youth League Branch Secretary (branch rated "Class A") 2024–2025 Outstanding Student Leader